Lab Assignment - 7

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1 Experiment 1: Limitations of Logistic Regression

1.1 Objective

The objective of this experiment is to study situations where Logistic Regression performs well and situations where it fails. Logistic Regression works best when the data is linearly separable but its performance degrades when the decision boundary is complex or non-linear.

1.2 Datasets

Three synthetic datasets were generated:

- **Dataset A:** Linearly separable dataset
- **Dataset B:** Dataset with overlapping classes
- **Dataset C:** Dataset with circular (non-linear) decision boundary

1.3 Methodology

For each dataset the following steps were performed:

- Logistic Regression model was trained
- Data points were visualized
- Decision boundary was plotted
- Classification accuracy was computed
- Confusion matrix was generated

1.4 Results

1.4.1 Dataset A

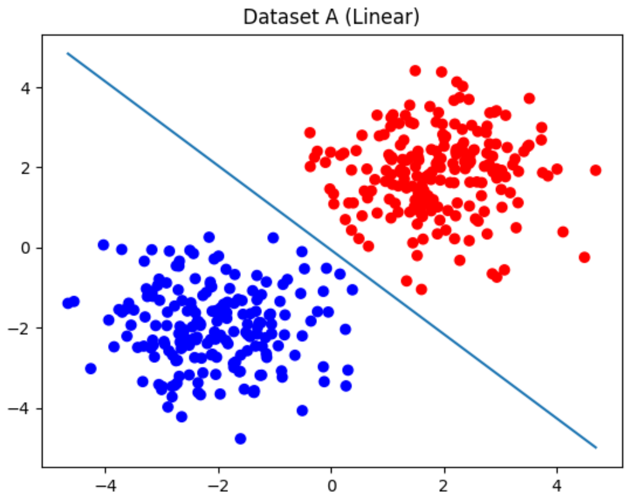

Figure 1: Decision Boundary for Linearly Separable Dataset

The classifier achieved high accuracy because the classes are clearly separable by a straight line.

1.4.2 Dataset B

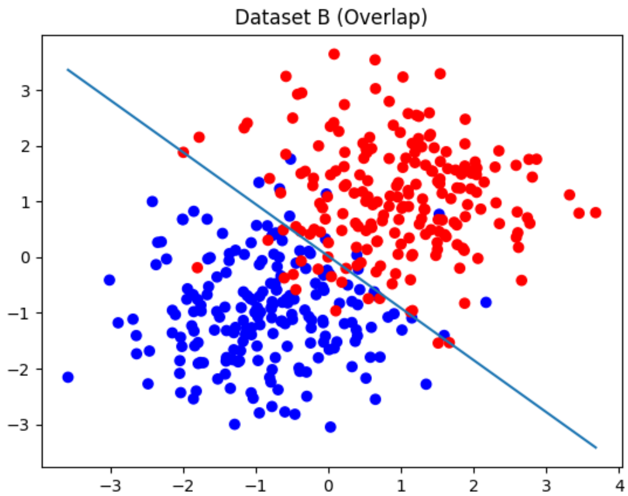

Figure 2: Decision Boundary for Overlapping Dataset

Accuracy decreased because samples from different classes overlap in feature space, making perfect separation impossible.

1.4.3 Dataset C

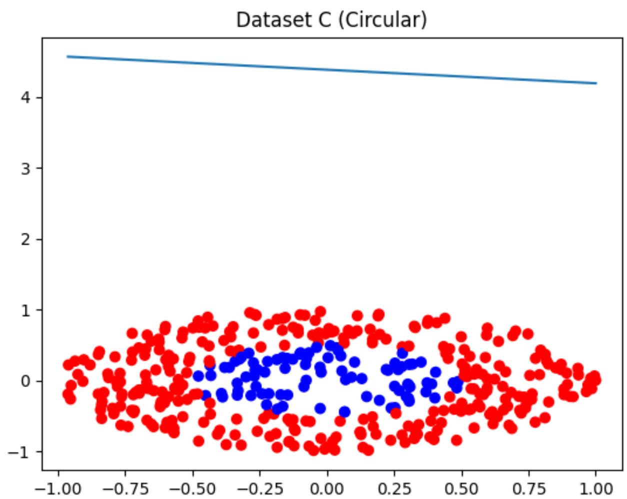

Figure 3: Decision Boundary for Circular Dataset

Performance was poor because logistic regression can only produce a linear decision boundary while the true boundary is circular.

1.5 Discussion

Why Logistic Regression performs well when the data is linearly separable

Logistic Regression models the probability of class membership as

$$P(y = 1 \mid x) = \sigma(w^T x + b)$$

where σ is the sigmoid function. The decision boundary is given by

$$w^T x + b = 0$$

which represents a linear boundary. If the dataset is linearly separable, the model can find parameters that perfectly separate the classes.

Why performance decreases when classes overlap

When class distributions overlap, samples from different classes share the same regions in feature space. In such situations no classifier can perfectly separate the data. Logistic Regression therefore produces a boundary that minimizes error but some misclassifications remain unavoidable.

Why Logistic Regression struggles with circular boundaries

Logistic Regression can only learn linear decision boundaries. In Dataset C the true boundary is circular and cannot be represented as a linear equation. Therefore the model fails to correctly classify many samples.

2 Experiment 2: Impact of Irrelevant Features

2.1 Objective

The objective of this experiment is to analyze how irrelevant or noisy features affect the performance of classification models.

2.2 Experimental Setup

A dataset was generated with two informative features that determine the class label. Additional random features were gradually added.

- Experiment 1: 2 informative features
- Experiment 2: 2 informative + 5 random features
- Experiment 3: 2 informative + 20 random features
- Experiment 4: 2 informative + 50 random features

Two classifiers were trained:

- Logistic Regression
- Gaussian Naive Bayes

The following metrics were recorded:

- Classification accuracy
- Training time
- Confusion matrix

2.3 Discussion

Effect of irrelevant features

Irrelevant features increase the dimensionality of the feature space without providing useful information about the target variable. This can lead to increased model variance and overfitting. The model may also learn spurious correlations between noise features and the target, which harms performance on unseen data.

Comparison of classifiers

Logistic Regression is a discriminative model that directly learns the relationship between features and the target variable. As the number of irrelevant features increases, the model must estimate more parameters, which increases variance and may reduce accuracy.

Gaussian Naive Bayes is a generative model that assumes conditional independence between features given the class label. Because each feature contributes independently to the likelihood, irrelevant features generally have less impact on the overall prediction. Therefore Naive Bayes often appears more robust to noisy features.

3 Experiment 3: Decision Boundary Analysis

3.1 Objective

The goal of this experiment is to analyze how dataset separability affects the performance of Logistic Regression trained using Gradient Descent.

3.2 Dataset Generation

Two datasets were generated with 500 samples each.

Dataset A (Linearly Separable)

$$z = 2x_1 + 3x_2 + \epsilon$$

Dataset B (Non-linear Boundary)

$$z = x_1^2 + x_2^2 - 1 + \epsilon$$

where

$$x_1, x_2 \sim U(-2, 2)$$

$$\epsilon \sim N(0, 0.2)$$

Class labels were defined as

$$y = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{otherwise} \end{cases}$$

3.3 Training Process

Logistic Regression was trained using Gradient Descent. During training the following values were recorded at every epoch:

- Training accuracy
- Testing accuracy
- Loss value

3.4 Plots

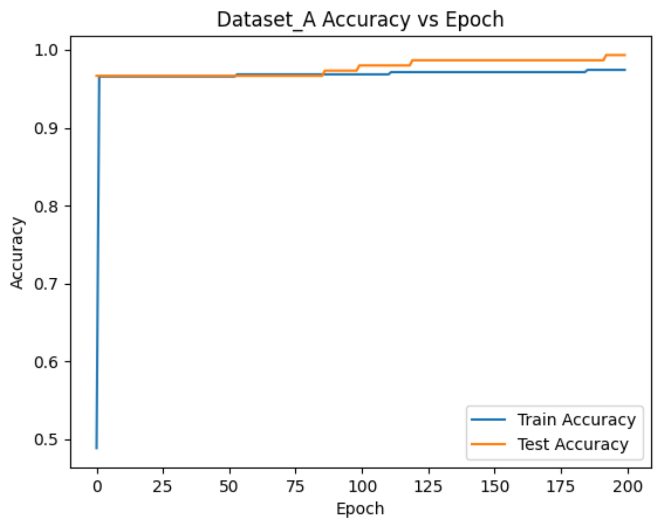

Figure 4: Accuracy vs Epoch for Dataset A

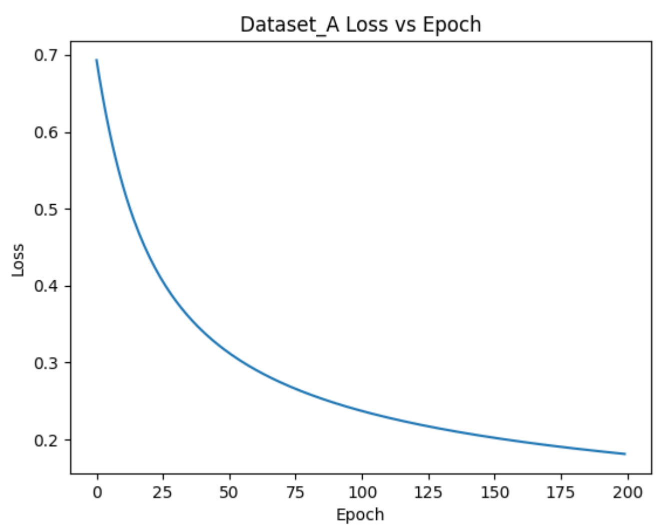

Figure 5: Loss vs Epoch for Dataset A

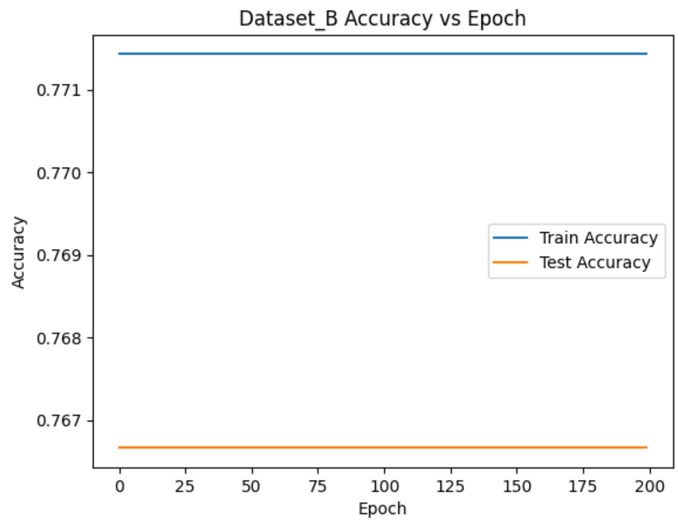

Figure 6: Accuracy vs Epoch for Dataset B

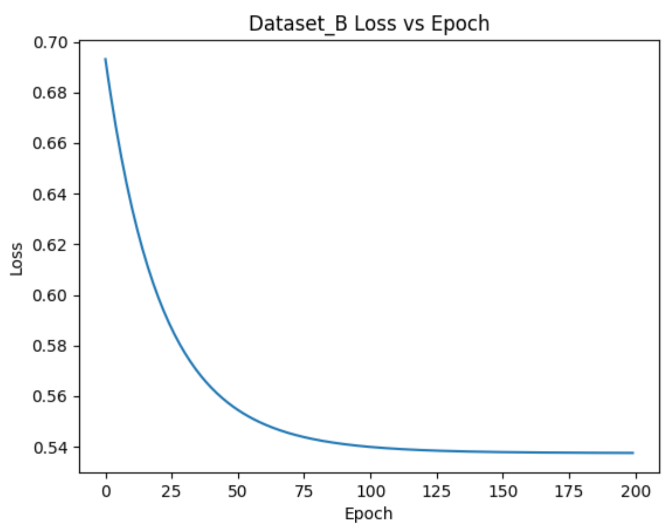

Figure 7: Loss vs Epoch for Dataset B

3.5 Analysis

Why Logistic Regression performs effectively for Dataset A

Dataset A is linearly separable, which matches the assumptions of Logistic Regression. As a result, the model quickly converges during training and achieves high accuracy.

Why the model struggles for Dataset B

Dataset B has a circular decision boundary. Logistic Regression can only learn linear boundaries, so it cannot accurately model the true structure of the data. As a result the model achieves lower accuracy and the loss stops improving after some epochs.

Alternative models
Several models can better capture non-linear boundaries:

- Support Vector Machines with non-linear kernels
- Neural Networks
- Decision Trees
- Random Forests

These models can learn complex decision surfaces that Logistic Regression cannot represent.